

Troubleshooting Guide: LPG Storage Sphere — Relief Valve Chatter from Thermal Relief Sizing Mismatch

Category: Storage & Relief Systems / Safety-Critical Equipment **Unit:** LPG Storage Sphere, Thermal (Fire/Solar) Relief PSV **Tools:** PSV thermal relief sizing review (API 521-based), sphere insulation/coating condition assessment **Fluid Package:** PR, used to determine LPG vapor pressure and relieving properties for thermal relief sizing calculations **Symptom:** A relief valve on an LPG storage sphere chattering (rapid open/close cycling) during normal hot-weather operation, without any abnormal process condition

1. Quick Reference

Item	Detail
Reported symptom	PSV on an LPG sphere chattering during normal, otherwise unremarkable hot-weather operation
Initially unclear	Whether the PSV itself was faulty/undersized, or whether the actual thermal relief load exceeded what the original sizing basis assumed
Actual root cause	Degraded sphere insulation/reflective coating had increased the actual solar heat absorption above the original design basis, so peak thermal relief load now exceeded the PSV's design relieving capacity margin, causing the valve to open right at (rather than safely above) its set condition and chatter
Fix	Restored/upgraded the sphere's reflective coating and insulation to reduce solar heat absorption back toward the original design assumption; reviewed PSV sizing margin against current conditions
Diagnostic signal	Chatter occurred specifically during peak solar loading conditions (clear, hot days), not during process upsets, pointing toward a thermal relief load issue rather than a process-driven overpressure or a purely mechanical PSV fault
Prevention	Periodic sphere coating/insulation condition assessment tied into PSV sizing basis review; PSV chatter logging correlated with weather conditions

2. Symptom

- **A PSV on an LPG storage sphere exhibited chatter** (rapid, repeated opening and reclosing) during **normal hot-weather operation** — not during any identifiable process upset.

3. Why This Wasn't Assumed to Be Simply a Faulty PSV

Chatter is often diagnosed as a mechanical valve problem — a worn seat, incorrect spring, or a valve significantly oversized for the actual relieving flow (a very common chatter cause, since an oversized valve reaches its full-open capacity almost immediately and reseats before adequate flow is achieved, cycling repeatedly). Before simply replacing or resizing the PSV, however, it was worth checking whether the underlying **relief scenario itself** — the actual thermal/fire relief load the valve was being asked to handle — still matched its original design basis, since sphere coating/insulation condition directly affects that load.

4. Diagnostic Approach

Step 1 — Correlate chatter events with weather/operating conditions

Chatter event timestamps were reviewed against weather data and process conditions.

Finding: Chatter occurred specifically during **clear, hot days with high solar loading** — not during any process upset, pressure excursion, or abnormal operating condition. This pointed toward a **thermal relief scenario** (solar/fire-case heat input) rather than a process-driven overpressure event.

Step 2 — Review the PSV's original thermal relief sizing basis

The PSV's original sizing calculation (API 521-based, using **PR** for LPG vapor pressure/relieving properties) was reviewed to confirm what solar heat absorption rate the thermal relief load was originally based on — typically assuming intact insulation and/or reflective coating per API 521 guidance.

Step 3 — Assess current sphere coating/insulation condition

Physical inspection of the sphere's exterior coating and insulation was performed, since the original sizing basis assumed a specific level of solar heat absorption that a coating in good condition helps limit.

Finding: The sphere's reflective coating/insulation had **degraded** (weathering, wear) since original installation, meaning actual solar heat absorption into the sphere was now **higher** than the original design assumption used for PSV thermal relief sizing.

Step 4 — Confirm the sizing margin mismatch

With degraded coating confirmed, the actual current thermal relief load (recalculated using PR and the higher effective solar absorption) was compared against the PSV's rated relieving capacity, confirming the **margin between required and available relief capacity had shrunk** — consistent with a valve chattering right at its opening/reseating threshold rather than lifting cleanly with adequate flow margin.

Quantitative Basis

- Sphere: 60 ft diameter, wetted surface area $A \approx 4,200 \text{ ft}^2$ to the design liquid level.

- API 521 fire/thermal relief heat input: $Q = 21,000 \cdot F \cdot A^{0.82}$. Original sizing basis assumed an environmental factor **F = 0.3** (credit for intact insulation/reflective coating): **$Q = 21,000 \times 0.3 \times 935 \approx 5.89$ MMBtu/hr.**
- PSV rated relieving capacity (in equivalent heat-input terms): **8.2 MMBtu/hr** — a design margin of 39% over the original 5.89 MMBtu/hr basis.
- With degraded coating, the effective environmental factor was reassessed at **F = 0.6** (roughly half the assumed insulation credit lost): **$Q = 21,000 \times 0.6 \times 935 \approx 11.78$ MMBtu/hr — 44% above the PSV's 8.2 MMBtu/hr rated capacity**, consistent with a valve unable to sustain stable full-open flow and cycling near its set pressure instead.

5. Root Cause

Degraded sphere insulation/reflective coating increased actual solar heat absorption above the original PSV thermal relief sizing basis, reducing the margin between the actual current thermal relief load and the valve's rated relieving capacity. This caused the PSV to open right at its set condition without achieving stable full flow, producing chatter — a sizing-margin issue driven by a physical condition change, not a fundamentally faulty valve.

6. Corrective Action

1. **Restored/upgraded the sphere's reflective coating and insulation**, reducing actual solar heat absorption back toward the original design assumption.
2. **Reviewed PSV sizing margin** against current (and going forward, periodically reassessed) conditions to confirm adequate relief capacity margin was restored.

7. Verification

- Post-restoration inspection confirmed coating condition consistent with **F = 0.3**, the original design basis.
- Recalculated thermal relief load returned to **5.89 MMBtu/hr — a restored 39% margin** below the PSV's 8.2 MMBtu/hr rated capacity.
- **No chatter observed over the following full summer season**, including multiple clear, high-solar-load days that had previously triggered the condition.

8. Prevention / Long-Term Fix

- Established **periodic sphere coating/insulation condition assessment tied into the PSV sizing basis review**, recognizing that PSV thermal relief sizing is only valid for as long as the assumed coating/insulation condition holds.
- Added **PSV chatter logging correlated with weather conditions**, so any future thermal-relief-related chatter is quickly distinguishable from process-driven relief events.

9. General Troubleshooting Checklist (Reusable)

- ☐ Correlate PSV chatter event timestamps against weather and process conditions before assuming a purely mechanical valve fault
- ☐ If chatter correlates with hot/high-solar-load conditions rather than process upsets, suspect a thermal (fire/solar) relief scenario
- ☐ Review the PSV's original thermal relief sizing basis (API 521-based) and its underlying assumptions about vessel coating/insulation condition

- ☐ Physically inspect current coating/insulation condition against that original assumption
- ☐ If degraded, recalculate the current actual thermal relief load and compare against the PSV's rated relieving capacity to confirm margin has shrunk
- ☐ Restore coating/insulation condition where practical, rather than defaulting straight to a PSV resize/replacement
- ☐ Re-verify PSV sizing margin after restoration
- ☐ Tie ongoing coating/insulation condition assessment into the PSV sizing basis review process, since sizing validity depends on maintaining the assumed vessel condition

10. Key Takeaway

PSV thermal relief sizing isn't a one-time calculation independent of vessel condition — it explicitly assumes a certain level of insulation/coating effectiveness limiting solar or fire-case heat input. When a relief valve starts chattering specifically during hot, high-solar-load conditions rather than process upsets, check whether the vessel's coating/insulation still matches the sizing basis before concluding the valve itself is faulty or needs resizing; restoring the coating can restore the intended relief margin without touching the valve at all.

Related Concepts / Tags

LPG-sphere PSV-chatter thermal-relief fire-case-relief API-521 solar-heat-absorption reflective-coating relief-valve-sizing Peng-Robinson

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.